

Release Plan

FreshBakes Bakery | IS501 Project

Methodology: Agile (Scrum) with Waterfall-style governance for Inventory & Payment modules **Team:** Two-person freelance development team **Total Duration:** 16 weeks (4 months) — targeting go-live before peak summer tourism season

Tooling

Purpose	Tool
Prototyping / Wireframes	Google Stitch / Figma
Diagrams (UML, DFD, ERD)	Mermaid (embedded in Markdown); Lucidchart for exports
Version control	Git
Project management	Jira / Trello (sprint board)
Payment integration	Stripe API
Frontend	React / Next.js
Backend	Node.js
Hosting	AWS / DigitalOcean (Cyprus-region)
Mobile delivery app	Progressive Web App (PWA)

Phase 1 — MVP Ordering (Weeks 1–8)

Goal: Replace notebooks and spreadsheets with a working digital ordering system.

Sprint	Weeks	Deliverables
Sprint 0	1–2	DB schema, cloud environment, Stripe webhook integration, inventory API (Waterfall-governed)
Sprint 1	3–4	Customer product browsing, order form, email confirmations
Sprint 2	5–6	Baker order queue dashboard, order status workflow
Sprint 3	7–8	Admin inventory management, low-stock alerts, basic admin dashboard

MVP Success Criteria

- Customer can place and pay for an order online ✓
 - Baker receives and processes the order digitally ✓
 - Inventory deducts automatically on payment ✓
 - Admin can manage stock and view orders ✓
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Phase 2 — Payments, Loyalty & Delivery (Weeks 9–16)

Goal: Add loyalty programme, delivery optimisation, and reporting — differentiate from competitors.

Sprint	Weeks	Deliverables
Sprint 4	9–10	Loyalty points system (earn + redeem), customer account history
Sprint 5	11–12	Mobile Delivery PWA — assignment, status updates, map integration
Sprint 6	13–14	Delivery route clustering, drag-to-reorder, estimated delivery times
Sprint 7	15–16	Sales & inventory reports, UAT, staff training, go-live

Phase 2 Success Criteria

- Loyalty programme active with 50+ enrolled customers ✓
 - Delivery staff operating exclusively via mobile app ✓
 - Admin generating weekly sales reports ✓
 - System live and handling real orders at full capacity ✓
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Module Methodology Mapping

Module	Methodology	Justification
Inventory management	Waterfall (Sprint 0)	Fixed, auditable rules; no UX iteration needed
Stripe payment integration	Waterfall (Sprint 0)	Regulatory and deterministic; tested against a written spec
Online ordering / UX	Agile Scrum	Customer-driven and evolving; needs real feedback each sprint
Custom cake configurator	Agile Scrum	Inherently iterative; high UX complexity with emergent requirements
Delivery route scheduling	Agile Scrum	Optimisation logic improves progressively with real usage data

Module	Methodology	Justification
Loyalty programme	Agile Scrum	Business rules evolve with promotion cycles and customer behaviour
Admin dashboard	Agile Scrum	Baker and admin needs best discovered through hands-on use

Delivery Route Optimisation — Iterative Sprint Breakdown

Demonstrating Agile's advantage over Waterfall for this specific module:

Sprint	Feature
Sprint 1	Drivers see assigned orders as cards; "Open in Google Maps" button per stop
Sprint 2	All stops plotted as pins on a single in-app map view
Sprint 3	Google Route Optimization API integrated; auto-generates fastest route
Sprint 4	Drag-and-drop reordering overlay; drivers can tweak sequence before starting

Key Milestones

Milestone	Target Week
Architecture sign-off & environment live	End of Week 2
First working order placed end-to-end	End of Week 4
Baker dashboard live in kitchen	End of Week 6
Stripe payments live	End of Week 8
Loyalty programme beta	End of Week 10
Delivery PWA in driver hands	End of Week 12
Full UAT complete	End of Week 15
Go-live	Week 16